Pareto-Optimal Sampling and Resource Allocation for Timely Communication in Shared-Spectrum Low-Altitude Networks

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***Abstract*—Guaranteeing stringent data freshness for low-altitude unmanned aerial vehicles (UAVs) in shared spectrum forces a critical trade-off between two operational costs: the UAV's own energy consumption and the occupation of terrestrial channel resources. The core challenge is to satisfy the aerial data freshness while finding a Pareto-optimal balance between these costs. Leveraging predictive channel models and predictive UAV trajectories, we formulate a bi-objective Pareto optimization problem over a long-term planning horizon to jointly optimize the sampling timing for aerial traffic and the power and spectrum allocation for fair coexistence. However, the problem's non-convex, mixed-integer nature renders classical methods incapable of fully characterizing the complete Pareto frontier. Notably, we show monotonicity properties of the frontier, building on which we transform the bi-objective problem into several single-objective problems. We then propose a new graph-based algorithm and prove that it can find the complete set of Pareto optima with low complexity, linear in the horizon and near-quadratic in the resource block (RB) budget. Numerical comparisons show that our approach meets the stringent timeliness requirement and achieves a six-fold reduction in RB utilization or a 6 dB energy saving compared to benchmarks.**

***Index Terms*—Pareto optimization, low-altitude networks, age of information (AoI), predictive communications.**

I. Introduction

Low-altitude activities have grown significantly over the past decade, resulting in a surge in demand for communication links that support time-critical applications, such as real-time navigation, control, and surveillance [1]. The core requirement for these services is not merely reliability, but stringent data freshness. For instance, the value of a UAV's field of view for mission monitoring is directly tied to its timeliness; a network that fails to deliver this information promptly renders it obsolete, severely compromising situational awareness and operational safety.

Most existing works on aerial communications focus on the throughput-reliability-delay trade-off [2], [3]. However, such approaches are insufficient to guarantee the stringent timeliness required by low-altitude networks. To address data freshness, some works employ AoI to assess the importance of information and timing and to control sampling and transmission decisions, thereby balancing efficiency and timeliness [4], [5]. Taking Figure 1 as an example, status-aware sampling control can simultaneously reduce power consumption and RB allocation while guaranteeing timeliness. However, the high mobility of UAVs creates a time-varying topology with rapidly changing channel conditions for both communication and interference links. This highly dynamic environment presents a significant challenge in jointly optimizing sampling and resource allocation.

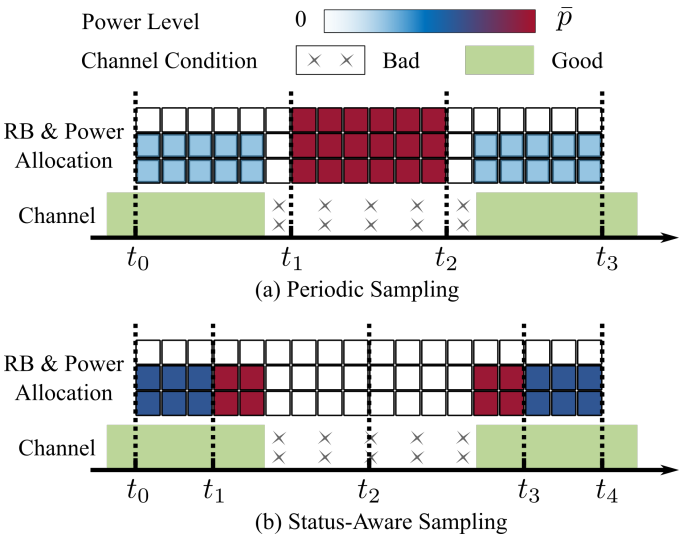


Figure 1. Illustration of periodic versus status-aware sampling policies. (a) A periodic policy enforces equal spacing and causes the update to fall within a poor-channel interval. (b) A status-aware policy adapts the sampling time to bypass the poor interval and satisfy the freshness constraint. Note that although status-aware sampling may involve more sampling and transmission instants, it exploits better channel intervals and therefore uses fewer RBs and less energy.

Low-altitude networks, while dynamic, are often predictable [6]. For instance, UAVs performing aerial inspection or delivery typically follow trajectories predetermined by their mission requirements, making their future channel conditions predictable. Some preliminary results have shown that this predictive information can enable proactive strategies to improve transmission efficiency, provided that a given data generation process is assumed [7], [8]. However, when data generation is optimized, the sampling timing and transmission strategy are tightly coupled, whereas exhaustive search for the best sampling timing is exponential in complexity.

In addition, the low-altitude and terrestrial networks are tightly coupled through dominant line-of-sight (LOS) channels, creating a direct conflict between their service objectives. Consequently, guaranteeing data freshness necessitates a trade-

off between two distinct operational costs: the UAV can either expend more of its own energy to increase transmission power, impacting its operational endurance, or it can occupy more of the shared channel resources to increase the spectrum for transmission, reducing its availability for terrestrial users. Therefore, a critical challenge is to find an optimal balance between these two costs, selecting a strategy that satisfies the timeliness requirement in the most efficient way for the coexistence of the two networks.

In this work, we formulate a bi-objective optimization problem to characterize the fundamental trade-off between aerial energy consumption and RB usage, under a strict constraint on aerial timeliness. The goal is to find the complete Pareto frontier of all optimal sampling and communication strategies. However, characterizing the complete Pareto frontier is a non-trivial task, as classical methods, such as weighted-sum approaches and heuristic algorithms, cannot guarantee finding all optimal trade-off points [9], [10]. This difficulty is compounded by the inherent complexity of the underlying joint optimization problem itself, which is non-convex, features an unknown number of variables (dimension-unknown), and involves mixed-integer constraints.

To tackle these challenges, we propose a predictive, status-aware framework with the following key contributions

- We propose a two-layer optimization framework, proven to find all Pareto optima, that first transforms the bi-objective problem into a single-objective equivalent and then decomposes it into an inner problem for communication strategy design and an outer problem for sampling control.
- A graph-based control algorithm is proposed to find the Pareto optimum with low complexity, linear in the horizon and near-quadratic in the RB budget.

II. System Model

Consider a UAV telemetry system, as depicted in Figure 2. The system consists of 1 UAV indexed by 0, and N base stations (BSs) indexed by $\mathcal{N} = \{1, \cdots, N\}$. The UAV is tasked with transmitting on-board sensory data to a fusion center via the terrestrial network (*i.e.*, the BSs). Timely delivery of this information is crucial, as the fusion center relies on the most recent and relevant data for reliable analysis and informed decision-making. Meanwhile, the terrestrial network shall also maintain stable service for the ground users.

To achieve fair coexistence of aerial and terrestrial traffic, we aim to design a resource management strategy that jointly determines (i) the optimal timing for sampling and transmission of aerial data, and (ii) the optimal power and spectrum allocation decisions. We leverage predictive channel models so that these decisions are optimal over a long-term planning horizon.

A. Predictive Channel Model

We consider a slotted orthogonal frequency division multiplexing (OFDM) system, where time is divided into slots indexed by $t \in \mathcal{T} = \{1, \cdots, T\}$, and the available spectrum

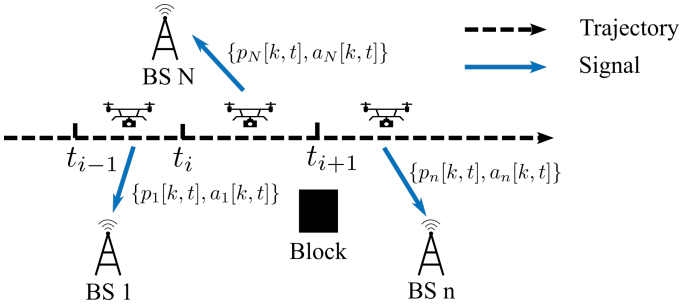


Figure 2. UAV telemetry system model. The UAV symbols along the trajectory illustrate the UAV's positions at different time instants. The UAV reports its state information or sensing data to a fusion center through BSs.

is partitioned into K orthogonal RBs, indexed by $k \in \mathcal{K} = \{1, \cdots, K\}$. Generally, the wireless channel from the UAV to BS n on RB k at time t can be modeled as

$$h_n[k,t] = g_n[k,t]\, \xi_n[k,t], \tag{1}$$

where $g_n[k,t]$ is the large-scale channel gain (e.g., path loss and shadowing), and $\xi_n[k,t] \sim \text{Gamma}(\kappa_n[k,t], 1/\kappa_n[k,t])$ captures small-scale fading with unit mean [8], [11].

The predictive channel model is built on two key enablers.

1) advanced channel sensing techniques, such as radio maps and digital twins [12], [13], which provide a 3D representation of the wireless propagation environment and offer spatially resolved channel statistics between the UAV and ground BSs; and
2) high-precision UAV control, which allows the UAV to follow pre-determined trajectories $\{(t, \mathbf{p}_0[t])\}_{t\in\mathcal{T}}$ with minimal deviation [6].

Consequently, the UAV's motion traces a one-dimensional slice through the 3D channel field, yielding a time-indexed channel profile that can be predicted in advance. Let

$$[g_n[k,t], \kappa_n[k,t]] = \Xi(\mathbf{p}_0[t], \mathbf{p}_n[t]), \ t \in \mathcal{T} \tag{2}$$

denote the radio map between the UAV and BS n along the trajectory. The resulting predictive channel model can then be expressed as

$$h_n[k,t] \sim \text{Gamma}(\kappa_n[k,t], g_n[k,t]/\kappa_n[k,t]). \tag{3}$$

B. Transmission Model

Denote the allocation of RB k at the time slot t for the communication from node 0 to BS n as $a_n[k,t] \in \{0,1\}$. For each (k,t) RB, the node 0 is allowed to transmit to at most one BS, that is

$$\sum_{n\in\mathcal{N}} a_n[k,t] \le 1, \ \forall k \in \mathcal{K}, t \in \mathcal{T}. \tag{4}$$

Let $p_n[k,t]$ denote the transmit power. The total transmission power is limited to the threshold $\bar{p}$, leading to the sum-power constraint

$$\sum_{k\in\mathcal{K}} \sum_{n\in\mathcal{N}} p_n[k,t] \le \bar{p}, \ \forall t \in \mathcal{T}. \tag{5}$$

We now derive the data throughput. The signal-to-noise ratio (SNR) for the link from node 0 to BS n on RB k at time t is given by $\gamma_n[k,t] = p_n[k,t]\, h_n[k,t]/\delta^2$, where δ^2 is the noise power. Assuming perfect Doppler compensation through advanced techniques [14], then, the channel capacity from node 0 to node n at time t for block k is modeled as

$$c_n[k,t] = B\log_2(1+\gamma_n[k,t]) \tag{6}$$

where B is the single-block bandwidth. Finally, the total data throughput over a time interval (t,t') aggregated across all receiving BSs and allocated RBs is given by

$$\upsilon(t,t') = \sum_{n\in\mathcal{N}}\sum_{k\in\mathcal{K}}\sum_{\tilde{t}=t}^{t'-1} c_n\left[k,\tilde{t}\right] a_n\left[k,\tilde{t}\right]. \tag{7}$$

C. Metrics for Coexistence

1) Timeliness requirement for aerial traffic: In this work, we use the AoI metric to quantify the freshness of information received from the UAV. Let $s[t]\in\{0,1\}$ denote the update-success indicator for node 0 at the end of the slot t, and t_0 denote its last sampling time. The AoI at the fusion center is recursively defined as

$$\tau[t+1] \triangleq \begin{cases} t-t_0, & s[t]=1,\\ \tau[t]+1, & s[t]=0. \end{cases} \tag{8}$$

A transmission attempt is successful if the expected delivered payload accumulated since the previous success meets a quality threshold $\bar{\upsilon}$, *i.e.*,

$$s[t] = \mathbb{I}\{\mathbb{E}\{\upsilon(t_0,t)\}\geq\bar{\upsilon}\}. \tag{9}$$

Here, the expectation is with respect to the channel $h_n[k,t]$ for all $n\in\mathcal{N}$, $k\in\mathcal{K}$ and $t\in[t_0,\cdots,t-1]$. For timeliness, we impose a hard constraint on the peak information age, *i.e.*,

$$\tau[t]\leq\bar{\tau},\ \forall t\in\mathcal{T}. \tag{10}$$

2) Fairness requirement for coexistence: For fairness between aerial and terrestrial services, we aim to regulate the aerial load so that no BS experiences excessive spectrum occupation by UAV traffic over time. To this end, we characterize the temporal load level at BS n by its worst-case load occupied by aerial traffic, i.e., $l_n \triangleq \max_{t\in\mathcal{T}}\sum_{k\in\mathcal{K}} a_n[k,t]$. To further promote spatial fairness across the network, we define the spatiotemporal load cap as

$$\theta \triangleq \max_{n\in\mathcal{N}} l_n = \max_{n\in\mathcal{N},t\in\mathcal{T}}\sum_{k\in\mathcal{K}} a_n[k,t]. \tag{11}$$

3) Energy efficiency for aerial traffic: In contrast to ground BSs, which have a stable and sufficient energy supply, the energy consumption at the UAV is tightly constrained by its limited onboard battery capacity, making energy efficiency a critical design consideration. Formally, we define the energy consumption as

$$E \triangleq \sum_{n\in\mathcal{N},k\in\mathcal{K},t\in\mathcal{T}} a_n[k,t]\, p_n[k,t]. \tag{12}$$

D. Age-Aware Sampling Controller

The AoI serves not merely as a timeliness constraint; it also allows direct control of information generation [5], [15], [16]. Given the predictive communication model over the entire planning horizon, we aim to control the sampling timing of sensory data so as to achieve our goal with minimal payload.

Mathematically, denote t_i as the sampling instant (*i.e.*, data generation time) of the ith status packet of the node 0. The sequence of sampling instants is $\mathbf{t}=\{t_1,t_2,\cdots,t_I\}$, where

$$1\leq t_i\leq T,\ t_i\in\mathcal{T}, i\in\mathcal{I}=\{1,\cdots,I\}, \tag{13}$$

and I is the total number of sampling events. Without loss of generality, we assume an initial sample is taken at the beginning of the first slot $t_0=1$.

Due to the AoI constraint (10), the sampling interval cannot be larger than $\bar{\tau}$, that is,

$$1\leq t_{i+1}-t_i\leq\bar{\tau},\ \forall i\in\mathcal{I}. \tag{14}$$

We set $t_{I+1}=T+1$ to represent the end time of transmission for the Ith status update. Recall that the transmission is successful if the link throughput satisfies (9). Accordingly, we impose the following expected throughput constraint between any two consecutive sampling instances

$$\mathbb{E}\{\upsilon(t_i,t_{i+1})\}\geq\bar{\upsilon},\ \forall i\in\mathcal{I}. \tag{15}$$

As a result, the AoI constraint (10) is equivalent to (13)-(15).

III. Problem Formulation

The goal is to maintain the timeliness of aerial data while satisfying coexistence requirements. To this end, we jointly optimize the sampling timing sequence $\mathbf{t}=\{t_1,t_2,\cdots,t_I\}$, transmit power $\mathbf{P}=\{p_n[k,t]\}_{n\in\mathcal{N},k\in\mathcal{K},t\in\mathcal{T}}$, and RB allocation $\mathbf{A}=\{a_n[k,t]\}_{n\in\mathcal{N},k\in\mathcal{K},t\in\mathcal{T}}$, to minimize the spatiotemporal load cap θ and the energy consumption E in the *Pareto* sense under hard AoI constraint. The bi-objective optimization problem is formulated as follows

$$\begin{aligned}\mathscr{P}1:\ &\underset{\mathbf{t},\mathbf{P},\mathbf{A}}{\text{minimize}}\ \{\theta,E\}\\ &\text{subject to } \mathbb{E}\{\upsilon(t_i,t_{i+1})\}\geq\bar{\upsilon},\forall i \qquad (16)\\ &\qquad\mathbf{t}\in\Upsilon,\mathbf{P}\in\mathcal{P}(\mathcal{T}),\mathbf{A}\in\mathcal{A}(\mathcal{T}). \qquad (17)\end{aligned}$$

Here, Υ, $\mathcal{P}$, and $\mathcal{A}$ denote the feasible spaces of sampling time, power allocation, and RB scheduling over the entire planning horizon $\mathcal{T}$, respectively. Specifically, from (13) and (15), the space of sampling time is

$$\Upsilon=\left\{\{t_i\in\mathcal{T}\}:1\leq t_{i+1}-t_i\leq\bar{\tau},\forall i\in\mathcal{I}(I),I\in\mathbb{R}^+\right\}.$$

From (5), the space of the power allocations is

$$\mathcal{P}(\mathcal{T})=\left\{\{p_n[k,t]\geq 0\}:\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{N}}p_n[k,t]\leq\bar{p},\forall t\in\mathcal{T}\right\},$$

and from (4), the RB allocation space $\mathcal{A}(\mathcal{T})$ is

$$\left\{\{a_n[k,t]\in\{0,1\}\}:\sum_{n\in\mathcal{N}}a_n[k,t]\leq 1,\forall k\in\mathcal{K},t\in\mathcal{T}\right\}.$$

The solution to $\mathscr{P}1$ in the Pareto sense is the determination of the complete Pareto frontier that consists of all Pareto-optimal strategies (see Definition 1 below). That is, there is no single 'best' strategy in Pareto optimization; instead, each point represents a trade-off between the two competing objectives.

Definition 1. (Pareto optimality.) A strategy $\pi = (\mathbf{t}, \mathcal{P}, \mathcal{A})$ is Pareto-optimal if there is no feasible strategy π' such that $\theta(\pi') \leq \theta(\pi)$ and $E(\pi') \leq E(\pi)$ with at least one strict inequality. The Pareto frontier is the set of all Pareto-optimal points in the objective space.

Remark: Problem $\mathscr{P}1$ presents several fundamental challenges: (i) determining the complete Pareto frontier is generally non-trivial, as illustrated in Figure 3; (ii) the underlying optimization is a non-convex mixed-integer problem, which is computationally demanding; and (iii) optimizing the variable $\mathbf{t}$ is particularly difficult since its dimension I is not fixed, leading to numerous isolated local optima. Mathematically, varying I corresponds to a mode-switching problem.

In the following sections, we shall derive some convenient properties of the Pareto frontier, building on which we develop an efficient graph-based method to solve the problem.

IV. Pareto Analysis and Problem Decomposition

This section characterizes the complete Pareto frontier. We first show that, for any given load cap θ, $\mathscr{P}1$ reduces to a single-objective problem $\mathscr{P}2$ that seeks to minimize the energy consumption $E^*(\theta)$ for a given θ. A strong result in Proposition 2 shows that $(\theta, E^*(\theta))$ in a specific domain characterizes the complete Pareto frontier. This result holds for any strictly increasing functions of the objectives θ and E. Moreover, $\mathscr{P}2$ can be decomposed into a two-layer problem that can be solved efficiently.

A. Pareto Analysis

Consider θ as an optimization variable, then the spatiotemporal load cap given by (11) can be equivalently represented by the following epigraph constraint [17]

$$\sum_{k\in\mathcal{K}} a_n[k,t] \leq \theta,\ \forall n \in \mathcal{N}, t \in \mathcal{T}. \tag{18}$$

Then, Problem $\mathscr{P}1$ reduces to a single-objective problem

$$\mathscr{P}2: \min_{\{\mathbf{t},\boldsymbol{\mathcal{P}},\boldsymbol{\mathcal{A}}\}\in\mathcal{F}(\theta)} \sum_{n\in\mathcal{N},k\in\mathcal{K},t\in\mathcal{T}} a_n[k,t]\, p_n[k,t],$$

where $\mathcal{F}(\theta)$ denotes the set of feasible strategies for a fixed load cap θ, where

$$\mathcal{F}(\theta) \triangleq \{\{\mathbf{t}, \boldsymbol{\mathcal{P}}, \boldsymbol{\mathcal{A}}\} : (16)\text{-}(18)\}. \tag{19}$$

Let $E^*(\theta)$ denote the optimal value of $\mathscr{P}2$. The following Proposition establishes the complete Pareto frontier of $\mathscr{P}1$.

Proposition 2. *(Pareto frontier.) The set*

$$\mathcal{C} \triangleq \{(\theta, E^*(\theta)) : \theta \in \{\underline{\theta}, \cdots \overline{\theta}\}\} \tag{20}$$

is the Pareto frontier of $\mathscr{P}1$, where $\underline{\theta} \triangleq \theta^*$,

$$\overline{\theta} \triangleq \min\{\theta \in \mathbb{Z}^+ : E^*(\theta) = E^*\}, \tag{21}$$

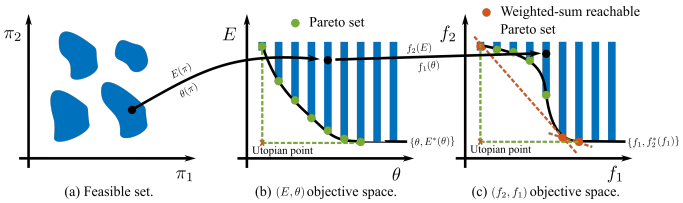


Figure 3. Illustration of Pareto optimality for a two-variable, two-objective optimization problem. (a) The nonconvex feasible set. (b) Bi-objective (E, θ) space, where the feasible space and the objective space are non-convex and non-continuous, thereby classical heuristic algorithms are challenging to find the complete Pareto frontier. (c) Bi-objective (f_2, f_1) space, where the Pareto frontier is non-concave, thereby, the weighted-sum method cannot guarantee the discovery of all Pareto optima.

and $\{\theta^*, E^*\}$ is the utopian point of the frontier (see Figure 3).

Proof: See Appendix A. ∎

This result holds for any strictly increasing scalarization of the bi-objective problem, which we formalize below.

Corollary 3. *For any $f_1(\cdot)$ and $f_2(\cdot)$, with $f_1'(\cdot) > 0$ and $f_2'(\cdot) > 0$. The set $\mathcal{C}_f \triangleq \{(f_1(\theta), f_2(E^*(\theta))) : \theta \in \{\underline{\theta}, \cdots \overline{\theta}\}\}$ is the Pareto frontier of the problem with the objective $\{f_1(\theta), f_2(E)\}$ under the same constraints as Problem $\mathscr{P}1$.*

Proof: See Appendix B. ∎

B. Problem Decomposition

It is observed from problem $\mathscr{P}2$ that the variables are coupled over time t only by the objective function and constraint (16). Therefore, given any feasible sampling variable $\mathbf{t}$, problem $\mathscr{P}2$ can be decomposed into I parallel resource allocation problems. Denote $\mathcal{T}_i \triangleq \{t_i, t_i+1, \cdots, t_{i+1}-1\}$ as the transmission interval of the ith sampling data, we can convert problem $\mathscr{P}2$ to a two-layer problem.

Proposition 4. *(Decomposition of $\mathscr{P}2$.) Problem $\mathscr{P}2$ is equivalently transformed into the following outer subproblem*

$$\mathscr{P}2\text{-}1: \quad \min_{\mathbf{t}} \sum_{i\in\mathcal{I}} E^*(t_i, t_{i+1}) \ \ s.t.\ \mathbf{t} \in \Upsilon,$$

where $E^*(t_i, t_{i+1})$ is the solution to the inner subproblem

$$\mathscr{P}2\text{-}2: \min_{\pi_i} \sum_{n\in\mathcal{N},k\in\mathcal{K},t=t_i}^{t_{i+1}-1} a_n[k,t]\, p_n[k,t]$$
$$s.t.\ (16)\ for\ i, \{p_m[k,t]\} \in \mathcal{P}(\mathcal{T}_i), \{a_m[k,t]\} \in \mathcal{A}(\mathcal{T}_i)$$

where $\pi_i = \{p_n[k,t], a_n[k,t]\}_{n\in\mathcal{N},k\in\mathcal{K},t\in\mathcal{T}_i}$.

Proof: See Appendix C. ∎

As a result, instead of solving the Problem $\mathscr{P}2$ directly, we can solve $\mathscr{P}2$-1 and $\mathscr{P}2$-2 separately, with less variable coupling.

V. Graph-Based Algorithm for Optimal Control

This section proposes a framework to solve Problem $\mathscr{P}2$-1 and Problem efficiently $\mathscr{P}2$-2.

First, $\mathscr{P}2$-2 can be transformed into a convex problem by relaxing $a_n[k,t] \in [0,1]$ and introducing an auxiliary variable

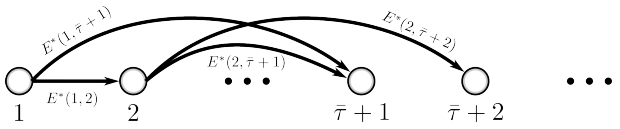


Figure 4. Illustration of the timing-control graph, where each vertex represents a possible sampling instant, each directed edge denotes a transmission during the interval between the two sampling instants, and the edge weight indicates the optimal energy consumption for the transmission induced by the edge.

Algorithm 1 Graph-based control algorithm

Input: θ, $g_n[k,t]$ and $\kappa_n[k,t]$

1) Construct the graph $\mathscr{G}$ based on Figure 4 and the weights are calculated by solving $\mathscr{P}$2-2 according to [8].
2) Shortest path algorithm from node 1 to node $T+1$ to find the optimal $\mathbf{t}^*$.
3) Calculate $\mathbf{P}_n^*$, $\mathbf{A}_n^*$, $E^*(\theta)$, according to $\mathbf{t}^*$.

Output: $\mathbf{P}_n^*$, $\mathbf{A}_n^*$, $\mathbf{t}^*$, $E^*(\theta)$

$\phi_n[k,t] = a_n[k,t]\, p_n[k,t]$. Consequently, $\mathscr{P}$2-2 can be solved with an optimality guarantee in $\mathcal{O}((t_{i+1}-t_i)\theta \log_2((t_{i+1}-t_i)\theta))$ complexity [8]. Thus, the remaining task is to solve the integer programming problem $\mathscr{P}$2-1.

A. Graph-Based Outer Solution

We construct a timing-control graph $\mathscr{G} = \{\mathbf{v}, \mathbf{e}, \mathbf{w}\}$, as shown in Figure 4. Here, the vertex set $\mathbf{v} = \{1, \cdots, T+1\}$ represents all admissible sampling timings; the terminal node $T+1$ marks the end boundary of the horizon so that each transmission interval is $\{t_i, t_i+1, \cdots, t_{i+1}-1\}$. The direct edge set $\mathbf{e} = \{(v_i, v_j)\}$ represents the transmission event, where $v_i, v_j \in \mathbf{v}$ for satisfying constraint $t_i \in \mathcal{T}$ and $1 \le t_i - t_j \le \bar{\tau}$ for satisfying AoI constraint. The weight set $\mathbf{w} = \{w_{i,j}\}$ represents the transmission cost, defining as the optimal solution to $\mathscr{P}$1-2 with sampling interval t_i and t_j, that is $w_{i,j} = E^*(t_i, t_j)$.

With this construction, a feasible sampling sequence $1 = t_0 < t_1 < \cdots < T_{I+1} = T+1$ corresponds to a directed path

$$1 \to t_1 \to \cdots \to T_{I+1} = T+1$$

whose total cost equals $\sum_{i=1}^{I} E^*(t_i, t_{i+1})$.

Proposition 5. *(Equivalence of $\mathscr{P}$2-1.) The optimal solution to $\mathscr{P}$2-1 is the shortest path from node 1 to node $T+1$ in $\mathscr{G}$.*

Proof: See Appendix D. ∎

Consequently, problem $\mathscr{P}$2-1 can be solved using a classical shortest path algorithm on a weighted, directed graph.

B. Graph-Based Algorithm

The algorithm is summarized in Algorithm 1, which proceeds in two phases: construct the timing-control graph $\mathscr{G}$ with edge weights given by the optimal interval energy $E^*(\cdot,\cdot)$; find the shortest path from node 1 to node $T+1$.

1) Optimality Analysis: First, the subproblem $\mathscr{P}$2-2 can be solved with an optimality guarantee as established in [8]. Therefore, each edge weight $w_{i,j}$ in the constructed graph $\mathscr{G}$ represents the optimal interval energy $E^*(t_i, t_j)$. Next, according to Proposition 5, the shortest path in $\mathscr{G}$ yields the optimal solution to the outer subproblem $\mathscr{P}$2-1. Finally, by Proposition 4, $\mathscr{P}$2-1 and $\mathscr{P}$2-2 together are equivalent to the original problem $\mathscr{P}$2. Accordingly, the output of Algorithm 1 achieves the globally optimal solution to $\mathscr{P}$2.

2) Complexity Analysis: The complexity for all Pareto frontier searching is $\mathcal{O}(T\theta^2\bar{\tau}^2 \log(\bar{\tau}\theta))$, where for each θ, the complexity for the shortest path over a directed graph with nonnegative weights is $\mathcal{O}(T\bar{\tau} + T) = \mathcal{O}(T\bar{\tau})$, and the complexity for calculating all the weights $E^*(t_i, t_j)$ in $\mathscr{G}$ is $\mathcal{O}(T\theta\bar{\tau}^2 \log \bar{\tau}\theta)$, which is proven in Appendix E.

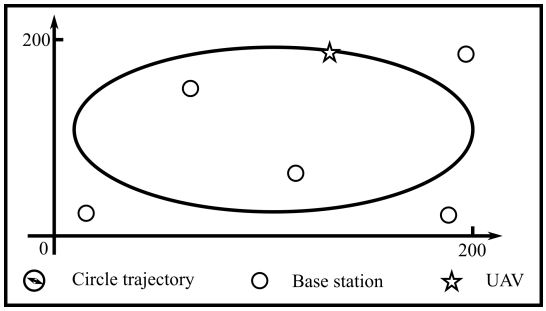


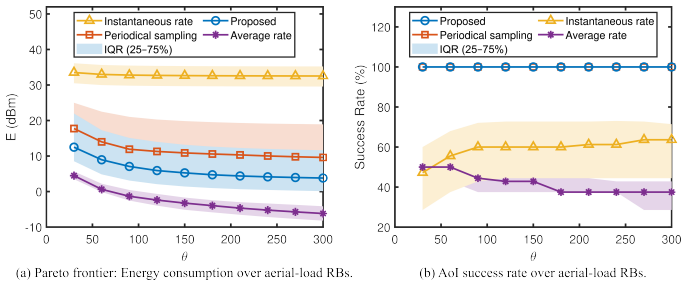
Figure 5. The simulation layout with $N = 5$ BSs and one patrol UAV.

VI. Simulation Results

We consider a UAV-based patrol system, where a UAV follows a circular trajectory to monitor a $200 \times 200\, m^2$ area, as illustrated in Figure 5. The UAV operates at an altitude of 50 m with a flight speed of 6 m/s. On the ground, N BSs are deployed, with their positions randomly generated. The channels are realized according to (1). Similar to [8], the shape parameters κ of the Gamma distribution of small-scale fading ξ are set randomly in [1, 30], and the large-scale fading g includes path loss and shadowing, where the path loss is generated by 3GPP Urban Micro (UMi) model [18] and the channel block state is generated by LOS probability model [19]. In contrast, the shadowing is modeled by a log-normal distribution.

We compare our performance with the following three baselines (two with no sampling control and one with sampling control but no sampling optimization). *1) Instantaneous rate [2]:* Trade-off spatiotemporal load cap and energy efficiency under the piecewise rate constraint, that is, $c_m(t) \ge S/\bar{\tau}$, $\forall t$. *2) Average rate [3]:* Trade-off spatiotemporal load cap and energy efficiency under the average rate constraint, that is, $\sum_t c_m(t)/T \ge S/\bar{\tau}$. *3) Periodical sampling:* The sampling time is fixed as $t_k = (k-1)\bar{\tau}$, while the resource allocation strategy follows the proposed schemes.

Figure 6 (a) illustrates the Pareto frontiers of all compared schemes. It can be observed that the non-predictive scheme (instantaneous rate) is significantly suboptimal, by more than



(a) Pareto frontier: Energy consumption over aerial-load RBs. (b) AoI success rate over aerial-load RBs.

Figure 6. Energy consumption and AoI success rate over the aerial-load RBs, where the solid lines with markers denote the median values, while the shaded areas indicate the interquartile range (IQR, 25th-75th percentile).

20 dB, compared with the other three predictive schemes, highlighting the importance of predictive information in strategy design. Moreover, although both the periodical and proposed status-aware schemes exploit predictive information, the latter achieves substantial performance improvement by adaptively controlling the sampling instants. For example, given an energy budget of $E = 10$ dBm, the periodical sampling method requires approximately 300 RBs to complete the task. In contrast, the proposed status-aware method needs only about 50 RBs.

Figure 6 (b) illustrates the fulfillment of the aerial timeliness requirement, showing that the achieved AoI remains below the maximum tolerable threshold. Firstly, although the average-rate method exhibits superior energy efficiency and RB utilization compared with other baselines, it fails to satisfy the AoI requirement consistently. This is because the average-rate predictive scheme optimizes the long-term average throughput rather than per-interval performance; thus, intervals coinciding with poor channel conditions remain underserved, leading to unstable AoI satisfaction even when θ increases. Similarly, the instantaneous rate method, despite consuming the most resources, also fails to guarantee the AoI constraint. Without prediction, it cannot anticipate deep fades; once a poor channel interval occurs, the per-interval requirement cannot be met. In contrast, the proposed predictive status-aware controller maintains 100% AoI satisfaction across all cases.

VII. Conclusion

In this paper, we designed a framework that jointly controls the data generation process (via sampling timing) and data transmission (via resource allocation). This approach successfully guarantees strict timeliness requirements while simultaneously minimizing both the UAV's energy consumption and its impact on the terrestrial network. The cornerstone of our approach is a novel two-layer graph-based algorithm, which we proved can efficiently characterize the complete Pareto frontier with a complexity of $\mathcal{O}(T\theta^2\bar{\tau}^2\log(\bar{\tau}\theta))$. The efficacy of this framework is demonstrated by significant resource savings, achieving up to a six-fold reduction in RB utilization and a 6 dB reduction in energy consumption in our simulations.

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APPENDIX A
PROOF OF PROPOSITION 2

We will prove that any point in $\mathcal{C}$ is Pareto-optimal and all Pareto-optimal solutions lie on $\mathcal{C}$. To establish this result, we first prove two key lemmas concerning the monotonicity of the $E^*(\theta)$.

A. Monotonicity of $E^*(\theta)$

Lemma 6. *For any $\theta_1 < \theta_2$, $E^*(\theta_1) \geq E^*(\theta_2)$.*

Proof: Given any $\theta_1 < \theta_2$. If $\mathcal{F}(\theta_1) = \emptyset$, we have $\mathcal{F}(\theta_1) \subseteq \mathcal{F}(\theta_2)$. Otherwise, for any $\mathcal{F}(\theta_1) \neq \emptyset$, for any $\{\mathbf{t}', \boldsymbol{\mathcal{P}}', \boldsymbol{\mathcal{A}}'\} \in \mathcal{F}(\theta_1)$, the load cap constraint (18) ensures $\sum_{k\in\mathcal{K}} a'_n[k,t] \leq \theta_1, \forall n, t$. Since $\theta_1 < \theta_2$, it follows that $\sum_{k\in\mathcal{K}} a'_n[k,t] \leq \theta_2, \forall n, t$. Moreover, all other constraints are independent of θ, the same tuple $\{\mathbf{t}', \boldsymbol{\mathcal{P}}', \boldsymbol{\mathcal{A}}'\}$ also satisfies (16) and (17). Therefore, $\{\mathbf{t}', \boldsymbol{\mathcal{P}}', \boldsymbol{\mathcal{A}}'\} \in \mathcal{F}(\theta_2)$, which establishes $\mathcal{F}(\theta_1) \subseteq \mathcal{F}(\theta_2)$.

With this result, minimizing the same objective E in $\mathscr{P}1$, over a larger feasible set cannot yield a higher optimum. Therefore,

$$E^*(\theta_1) \geq E^*(\theta_2), \forall \theta_1 < \theta_2$$

showing that $E^*(\theta)$ is non-increasing in θ. ∎

Lemma 7. *For any $\theta_1 < \theta_2 \in \{\underline{\theta}, \cdots, \overline{\theta}\}$, $E(\theta_1) > E(\theta_2)$.*

Proof: Suppose, to the contrary, that there exist $\underline{\theta} \leq \theta_1 < \theta_2 \leq \overline{\theta}$ such that $E^*(\theta_1) = E^*(\theta_2)$. Let $\{\mathbf{t}', \boldsymbol{\mathcal{P}}', \boldsymbol{\mathcal{A}}'\} \in \mathcal{F}(\theta_1)$ be any optimal solution of problem $\mathscr{P}2$ over $\mathcal{F}(\theta_1)$. Because $\mathcal{F}(\theta_1) \subseteq \mathcal{F}(\theta_2)$ due to $\theta_1 < \theta_2$ and $E^*(\theta_1) = E^*(\theta_2)$ hold, $\{\mathbf{t}', \boldsymbol{\mathcal{P}}', \boldsymbol{\mathcal{A}}'\}$ is also optimal for $\mathscr{P}2$ over $\mathcal{F}(\theta_2)$.

By the definition of $\mathcal{F}(\theta)$ in (19), its per-(n,t) RB constraints satisfy $\sum_k a'_n[k,t] \leq \theta_1, \forall n, t$. Hence, we have $\sum_k a'_n[k,t] \leq \theta_1 < \theta_2, \forall n, t$. That is, all RB-cap constraints are inactive (strictly slack) at the θ_2-problem realized by $\boldsymbol{\mathcal{A}}'$.

From parametric sensitivity analysis (see [17]), the derivative of the optimal value with respect to a right-hand-side parameter equals the optimal Lagrange multiplier of the corresponding constraint. If the constraint is inactive (strictly slack), its multiplier is zero, and any right-hand-side relaxations do not affect the optimal value or the primal minimizers. Hence, the same solution remains optimal for all $\theta_3 > \theta_2$, which implies

$$E^*(\theta_3) = E^*(\theta_2) = E^*(\theta_1), \forall \theta_3 \in \{\theta_2 + 1, \cdots, \overline{\theta}\}$$

In particular, evaluating at $\theta_3 = \overline{\theta}$ yields $E^* = E^*(\overline{\theta}) = E^*(\theta_1)$, which contradicts the definition of $\overline{\theta}$ in (21), where $\overline{\theta}$ is the minimal θ such that $E^* = E^*(\theta)$.

Therefore, the assumption is false, and we conclude that $E^*(\theta_1) > E^*(\theta_2)$ for any $\underline{\theta} \leq \theta_1 < \theta_2 \leq \overline{\theta}$. ∎

B. Points on $\mathcal{C}$ are Pareto-optimal.

For any $\theta \in \{\underline{\theta}, \cdots, \overline{\theta}\}$, suppose, to the contrary, there exists a feasible pair (θ', E') that weakly dominates $(\theta, E^*(\theta))$, *i.e.*,

$$\theta' \leq \theta, E' \leq E^*(\theta), \text{ and } (\theta, E^*(\theta)) \neq (\theta', E').$$

By definition of $E^*(\cdot)$, feasibility at θ' implies $E' \geq E^*(\theta')$. Hence

$$E^*(\theta') \leq E' \leq E^*(\theta).$$

If $\theta' < \theta$, strictly monotonicity on $\{\underline{\theta}, \cdots, \overline{\theta}\}$ according to Lemma 7 gives $E(\theta') > E(\theta)$, which contradicts $E(\theta') \leq E(\theta)$. If $\theta' = \theta$, then $E(\theta') = E(\theta)$ force $E' = E(\theta)$, contradicting $(\theta, E(\theta)) \neq (\theta', E')$.

Thus, no feasible pair weakly dominates $(\theta, E^*(\theta))$; hence, any $(\theta, E^*(\theta)) \in \mathcal{C}$ is Pareto-optimal.

C. Every Pareto-optimal feasible pair lies on $\mathcal{C}$.

Let (θ', E') be any feasible pair. By definition of $E^*(\cdot)$, $E' \geq E^*(\theta')$. If $E' > E^*(\theta')$, then $(\theta', E^*(\theta'))$ (which is feasible) strictly improves the energy objective without worsening θ, so (θ', E') is dominated and cannot be Pareto-optimal. Therefore any Pareto-optimal feasible pair must satisfy $E' = E^*(\theta')$, *i.e.*, it lies on $\mathcal{C}$.

APPENDIX B
PROOF OF COROLLARY 3

Since $E^*(\theta)$ is non-increasing (strictly decreasing on $\{\underline{\theta}, \cdots \overline{\theta}\}$), and $f_1(\cdot)$ and $f_2(\cdot)$ are strictly increasing, the composition $f_2 \circ E^* \circ f_1$ preserves order and strictness on that interval. The mapping $(\theta, E) \mapsto (f_1(\theta), f_2(E))$ is thus order-preserving, hence the image of $\mathcal{C}$ is again a Pareto frontier.

APPENDIX C
PROOF OF PROPOSITION 4

Separate the energy consumption over status update, we have $E = \sum_i \sum_{n,k,t=t_i}^{t_{i+1}-1} a_n[k,t]\, p_n[k,t]$. Then, given $\mathbf{t}$, $\mathscr{P}2$ becomes

$$\min_{\mathbf{P}\in\mathcal{P}(\mathcal{T}), \mathbf{A}\in\mathcal{A}(\mathcal{T})} \sum_i \sum_{n,k,t=t_i}^{t_{i+1}-1} a_n[k,t]\, p_n[k,t] \text{ s.t. (16, 18)}$$

where the constraint for $\mathbf{t}$ is removed because $\mathbf{t}$ is given, which will be optimized later.

Denote $\pi_i = \{p_n[k,t], a_n[k,t]\}_{n\in\mathcal{N}, k\in\mathcal{K}, t\in\mathcal{T}_i}$, where $\mathcal{T}_i = \{t_i, t_i+1, \cdots, t_{i+1}-1\}$, it is observed the variables for $\mathbf{x}_i$ and π_j for $i \neq j \in \mathcal{I}$ are uncoupled. Therefore, problem $\mathscr{P}2$ given $\mathbf{t}$ can be written as

$$\sum_i \min_{\pi_i} \sum_{n,k,t=t_i}^{t_{i+1}-1} a_n[k,t]\, p_n[k,t]$$
$$\text{s.t.(16) for } i, \{p_m[k,t]\} \in \mathcal{P}(\mathcal{T}_i), \{a_m[k,t]\} \in \mathcal{A}(\mathcal{T}_i).$$

Denote the optimal value to each transmission as $E^*(t_i, t_{i+1})$, then, Problem $\mathscr{P}2$ can be written as

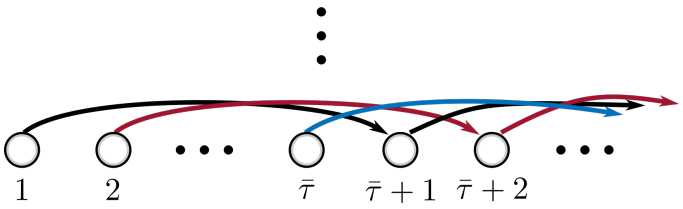
$$\min_{\mathbf{t}} \sum_i E^*(t_i, t_{i+1}) \text{ s.t. } \mathbf{t} \in \Upsilon.$$



(a) Edges with c = 1; There exists one path, from 1 to T+1.



(b) Edges with c = 2; There exists two paths, from 1 to T (black), and from 2 to T+1 (red).



(c) Edges with c = $\bar{\tau}$; There exist $\bar{\tau}$ paths, from 1 to T-$\bar{\tau}$ (black), ..., and from $\bar{\tau}$ to T+1 (blue).

Figure 7. Edge division according to transmission interval lengths, *i.e.*, $t_j - t_i$.

Appendix D
Proof of Proposition 5

Path-feasible Correspondence. Feasible $\{t_i\}$ in $\mathscr{P}$2-1 satisfies $1 \le t_{i+1} - t_i \le \bar{\tau}, \forall i$ and $1 = t_0 \le \cdots \le t_{I+1} = T, t_i \in \mathcal{T}, \forall i$. By graph construction, each consecutive pair (t_i, t_{i+1}) is a valid edge; hence $\{t_i\}$ induces a path $1 \to t_1 \to \cdots \to T_{I+1} = T + 1$. Vice versa.

Cost equivalence. Each edge (t_i, t_{i+1}) has weight $w_{i,i+1} = E^*(t_i, t_{i+1})$, thus, the path length $\sum_i w_{i,i+1}$ equals the outer objective $\sum_i E^*(t_i, t_{i+1})$ at the corresponding $\{t_i\}$.

Accordingly, minimizing $\sum_i E^*(t_i, t_{i+1})$ over feasible $\{t_i\}$ is identical to finding the shortest $1 \to T + 1$ in $\mathscr{G}$.

Appendix E
Complexity Analysis for Graph $\mathscr{G}$ Construction

To analyze the computational cost of constructing $\mathscr{G}$, we partition the edges into $\bar{\tau}$ groups indexed by $c \in 1, \cdots, \bar{\tau}$; in group c, every edge (v_i, v_j) induces a transmission interval of length c, that is, $v_j - v_i = c$, as shown in Figure 7. Since any valid edge in $\mathscr{G}$ must satisfy $1 \le t_i - t_j \le \bar{\tau}$ according to the construction policy described in Section V-A, the union of all $\bar{\tau}$ groups collectively constitutes the complete edge set of $\mathscr{G}$.

For each group, there are c paths from near 1 to near $T+1$, and the cost for calculating the weight of edges in this group is less than $cT\theta \log_2(c\theta)$. For example, when $c = 1$, there is only one path $1 \to 2 \to 3 \to \cdots \to T+1$ and the transmission interval 1. Therefore, the complexity is

$$\sum_i^T (t_{i+1} - t_i)\theta \log_2((t_{i+1} - t_i)\theta) = \sum_i^T \theta \log_2(\theta) = T \log_2(\theta).$$

When $c = 2$, there is only two paths $1 \to 3 \to 5 \to \cdots \to T$ and $2 \to 4 \to 6 \to \cdots \to T + 1$, and the transmission interval is 2. Therefore, the complexity is

$$2 \sum_i^{\lfloor T/2 \rfloor} (t_{i+1} - t_i)\theta \log_2((t_{i+1} - t_i)\theta) = 2 \sum_i^{\lfloor T/2 \rfloor} 2\theta \log_2(2\theta) \le 2T \log_2(\theta).$$

Following the same reasoning, the computational cost for groups with $c \in 3, \cdots, \bar{\tau}$ can be derived analogously.

As a result, the complexity for calculating all weights in the graph $\mathscr{G}$ is less than

$$\begin{aligned}
&\sum_{c=1}^{\bar{\tau}} cT\theta \log_2(c\theta) = T\theta \sum_{c=1}^{\bar{\tau}} c \log(c) + T\theta \sum_{c=1}^{\bar{\tau}} c \log(\theta) \\
&= T\theta \sum_{c=1}^{\bar{\tau}} c \log(c) + T\theta \log(\theta) \sum_{c=1}^{\bar{\tau}} c \\
&= T\theta \sum_{c=1}^{\bar{\tau}} c \log(c) + T\theta \log(\theta) \frac{\bar{\tau}(\bar{\tau}+1)}{2} \\
&\le T\theta \int_{c=1}^{\bar{\tau}} c \log(c)\, dc + T\theta \log(\theta) \frac{\bar{\tau}(\bar{\tau}+1)}{2} \\
&= T\theta \left(\frac{\bar{\tau}^2}{2} \log_2(\bar{\tau}) - \frac{\bar{\tau}^2}{4\ln(2)} + \frac{1}{4\ln(2)} \right) \\
&\qquad + T\theta \log(\theta) \frac{\bar{\tau}(\bar{\tau}+1)}{2} \\
&= \mathcal{O}\left(T\theta\bar{\tau}^2 \log(\bar{\tau}\theta)\right)
\end{aligned}$$

which is the order of

$$\mathcal{O}\left(T\theta\bar{\tau}^2 \log(\bar{\tau})\right) + \mathcal{O}\left(T\theta\bar{\tau}^2 \log(\theta)\right) = \mathcal{O}\left(T\theta\bar{\tau}^2 \log(\bar{\tau}\theta)\right).$$

In summary, the complexity for constructing graph $\mathscr{G}$ is $\mathcal{O}\left(T\theta\bar{\tau}^2 \log(\bar{\tau}\theta)\right)$.